

Dr. Priyanka Venkatesan

Quantitative Analyst — Market Risk

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SUMMARY

Market-risk quant with a PhD in applied mathematics from IIT Bombay and 4 years developing and defending internal models at a private-sector bank's treasury. Owned ES/VaR methodology through the FRTB parallel-run under RBI's draft guidelines, including P&L attribution testing and NMRF treatment, plus the IRRBB EVE/NII engine. Comfortable in front of Model Validation and RBI inspection teams. Notice period: 60 days.

SKILLS

Quant: Expected Shortfall/VaR, FRTB (PLA tests, NMRF), risk-factor modeling, backtesting, Monte Carlo methods, G-Sec/OIS curve construction, vanilla-option pricing and Greeks, IRRBB (EVE/NII per RBI circular)

Code: Python (NumPy/SciPy/pandas, production risk libraries), C++ (contributor level), SQL, Git

Governance: Model documentation, validation responses, RBI inspection Q&A support, FIMMDA valuation norms

EXPERIENCE

Quantitative Analyst, Market Risk — Vindhya Bank, Mumbai

Aug 2022 – present

- Co-developed the bank's Expected Shortfall model for the FRTB parallel-run (rates & FX trading books); P&L attribution tests passing in 11 of 12 months on the pilot desks.
- Redesigned risk-factor bucketing for non-modellable risk factors, reducing the projected NMRF capital add-on by 18% with documented statistical justification.
- Built the backtesting framework (Python) now running daily against 3 years of history; investigated and closed 100% of VaR breaches within SLA.
- Rebuilt the IRRBB EVE/NII scenario engine to the RBI prescribed shocks; presented results to ALCO monthly.
- Supported two RBI inspection cycles: methodology walkthroughs and data lineage evidence.

Quantitative Analyst (rotational) — Vindhya Bank, Mumbai

Sep 2021 – Jul 2022

- Rotations across counterparty risk (SA-CCR implementation checks) and treasury mid-office analytics.

EDUCATION

PhD Applied Mathematics (stochastic analysis) — IIT Bombay, 2017 – 2021

M.Sc. Applied Mathematics — Anna University, Chennai, 2015 – 2017

B.Sc. Mathematics — Loyola College, Chennai, 2012 – 2015

PUBLICATIONS

"A copula approach to aggregating non-modellable risk factors" — Journal of Risk, 2023.

CERTIFICATIONS

FRM (2023)

LANGUAGES KNOWN

Tamil (native), English (fluent), Hindi (fluent)